

Homework 7

Due March 22nd

Problems:

- Two amplifier circuits, one an inverting amplifier with $A_{0,1}$ and one non-inverting amplifier with $A_{0,2}$, are cascaded to produce an overall circuit with a DC gain of 100 V/V (applicable to each subquestion). Both subcircuits use the same op amp, which has an open-loop gain of 500 V/mV and a gain bandwidth product of 15 MHz.
 - If both amplifier subcircuits have the same [closed-loop] bandwidth, what must be the bandwidth of the full circuit? Further, what is the gain of each subcircuit? Hint: what is the relationship between gain and feedback factors?
 - If both amplifier subcircuits have the same gain magnitude, what is the bandwidth of each subcircuit and the full circuit? Hint: it should be very similar to your answer from the previous problem.
 - If $A_{0,1} = -50$ V/V, what is the bandwidth of each subcircuit and the full circuit?
- An inverting amplifier circuit is configured to provide a DC gain of $A_0 = -5$ V/V using an op amp with a gain bandwidth product of 1 MHz.
 - Find the **minimum slew rate** required for the op amp such that the amplifier circuit is ensured to be **small signal bandwidth limited**; that is, slew rate limiting does not reduce the effective bandwidth of the circuit below f_B . Assume the amplifier is powered by ± 5 V supplies and is rail-to-rail.
 - Assuming an input signal of $v_i = 0.625 \sin(2 \times 10^5 t + 75^\circ)$ V, what is the minimum slew rate to prevent slew rate limiting of the output?
 - Simulate the circuit in LTspice with the voltage input signal from (b). Pick realistic values for the resistors to produce the given gain. For the op amp: use UniversalOpAmp2, which is a simplistic op amp model which includes slew rate. You'll need to edit the amplifier (right-click) to modify the GBP (GBW in LTspice) and slew rate (unit: V/s). Evaluate the simulation using the slew rate calculated for (b) and at values 25%, 50%, and 75% of this value. For each, plot the output voltage against the ideal output voltage. To get the ideal output voltage: either modify the trace of the input voltage (multiply by the gain) or produce another voltage source that outputs the ideal output.
 - It is desired to use the inverting amplifier circuit above (unmodified) as a component of a larger circuit to provide an effective gain of $A_0 = 75$ V/V. Design a cascaded circuit to produce this gain using only resistors and the op amp as given above (use as many as needed). What is the effective bandwidth and closed-loop gain bandwidth product of the resulting circuit?
- Book Problem 6.35.
- Show that a voltage follower based on a current feedback amplifier with feedback resistor $R = 10$ k Ω (shown below), does indeed provide an ideal gain of $A_{ideal} = 1$. Further, with the internal components of $z(jf)$ known as $R_{eq} = 500$ k Ω and $C_{eq} = 2$ pF, what is the gain error of the circuit at $f = 0$ Hz, 100 kHz, 1 MHz, and 10 MHz?

